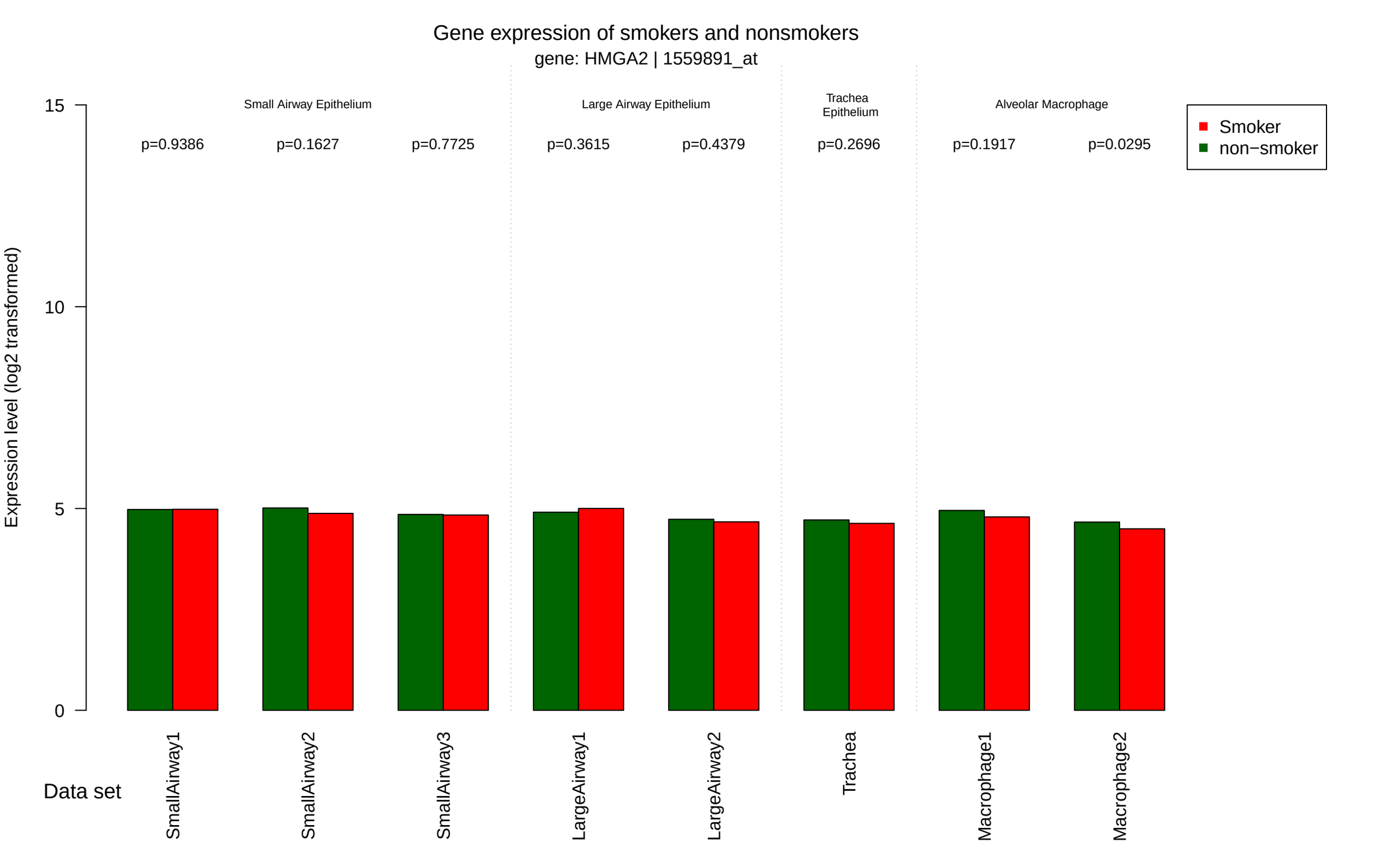
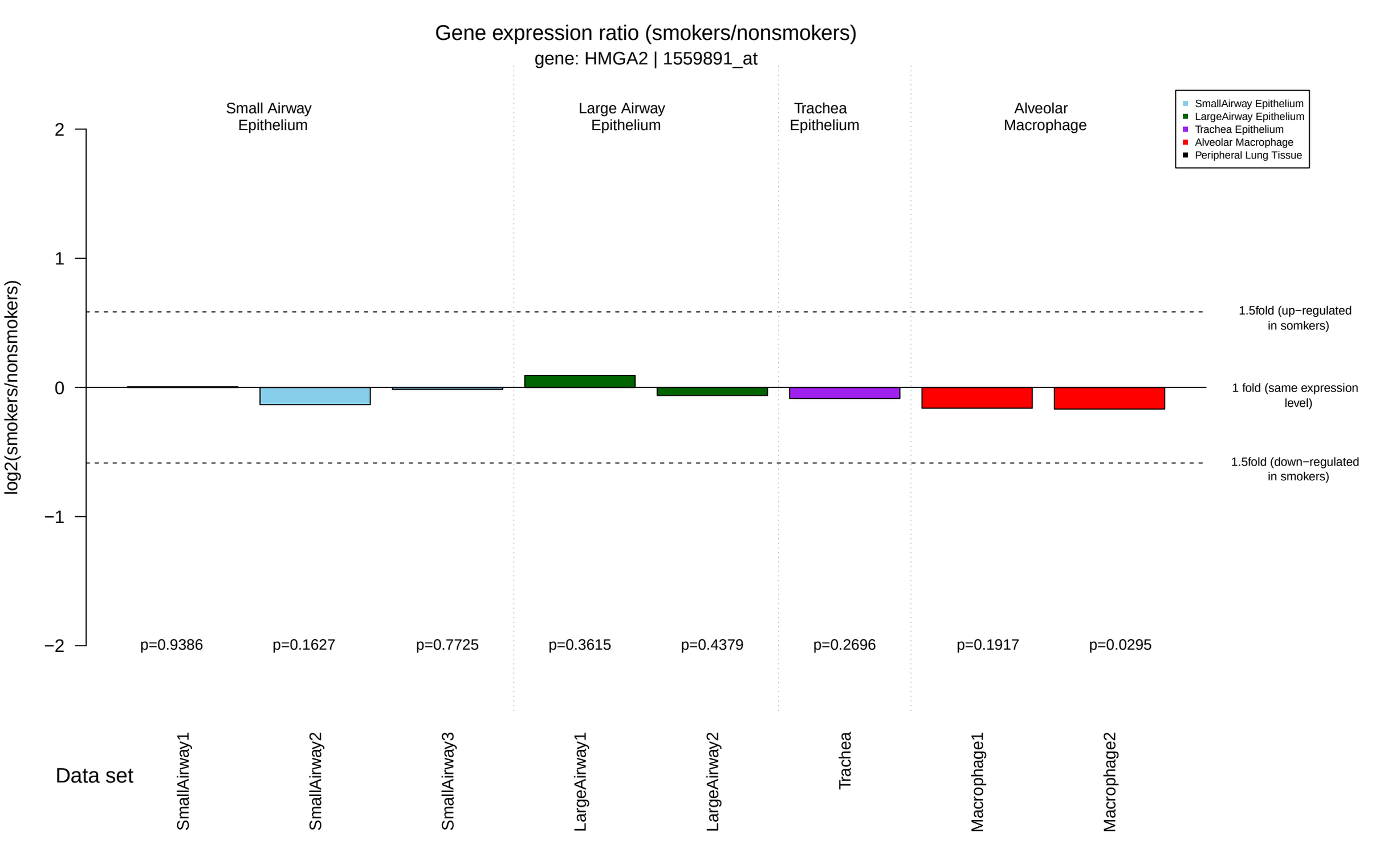
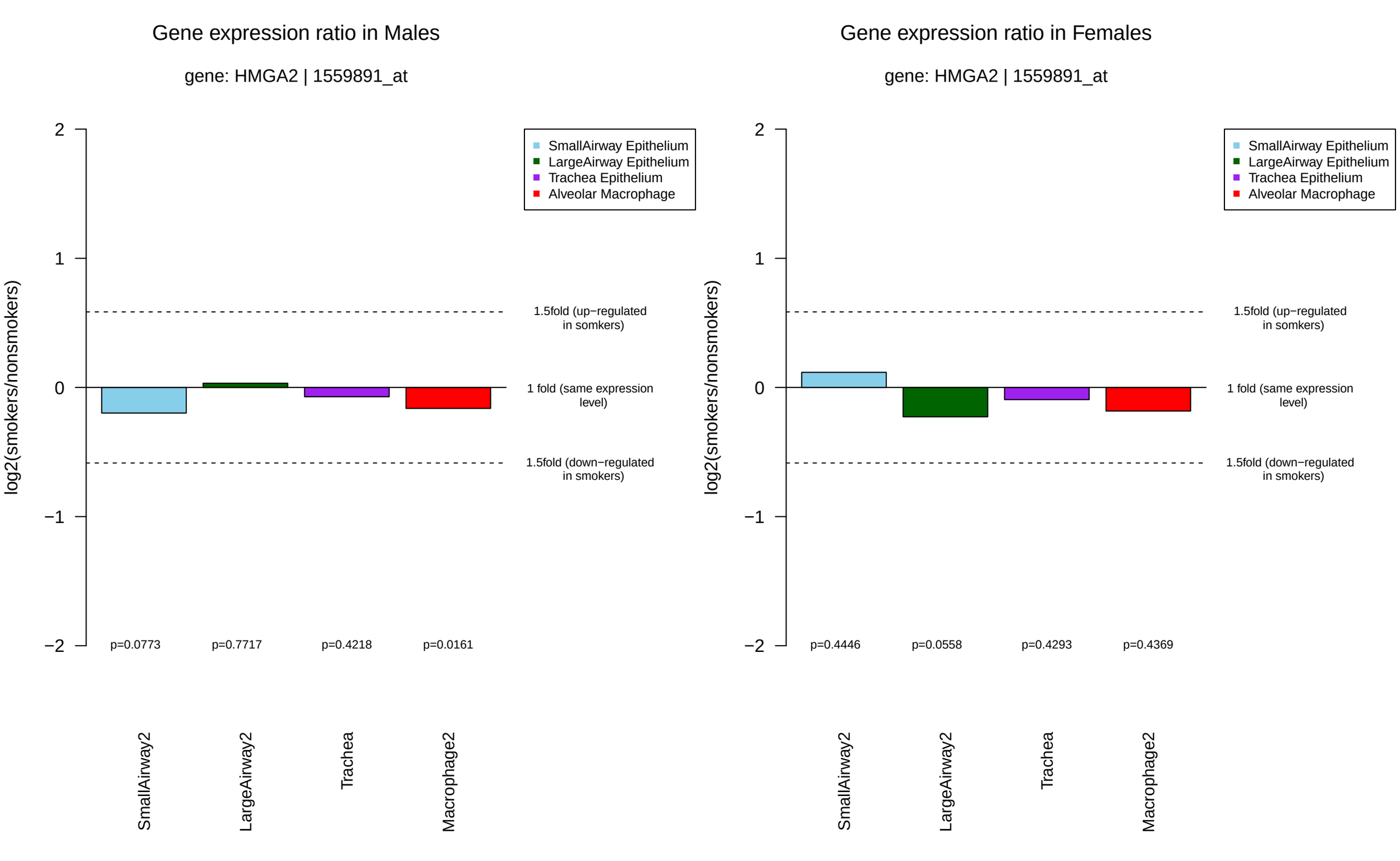
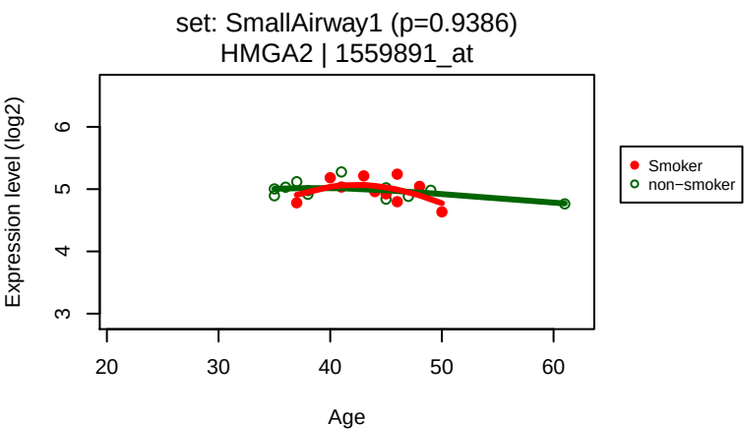
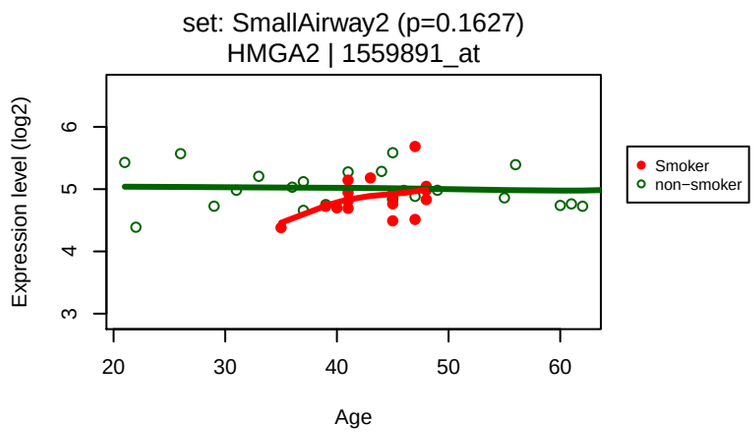
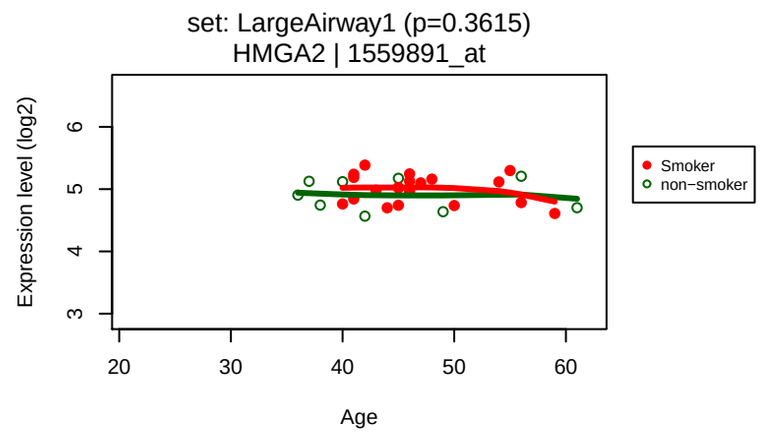

Gene expression of smokers and nonsmokers
gene: HMGA2 | 1559891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9386
p=0.1627
p=0.7725
p=0.3615
p=0.4379
p=0.2696
p=0.1917
p=0.0295
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

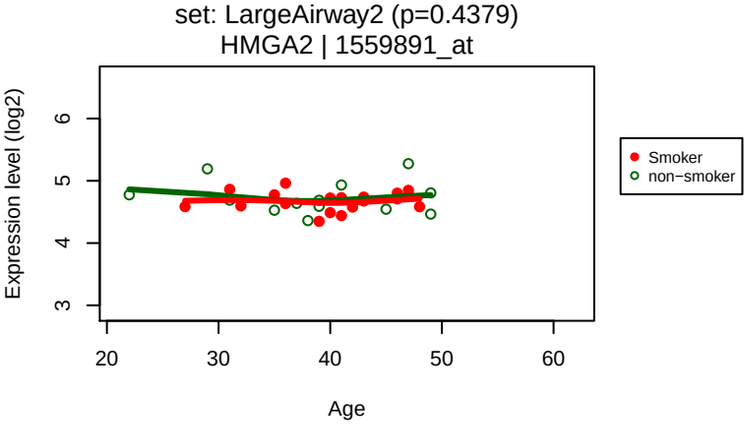

Gene expression ratio (smokers/nonsmokers)
gene: HMGA2 | 1559891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9386
p=0.1627
p=0.7725
p=0.3615
p=0.4379
p=0.2696
p=0.1917
p=0.0295
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

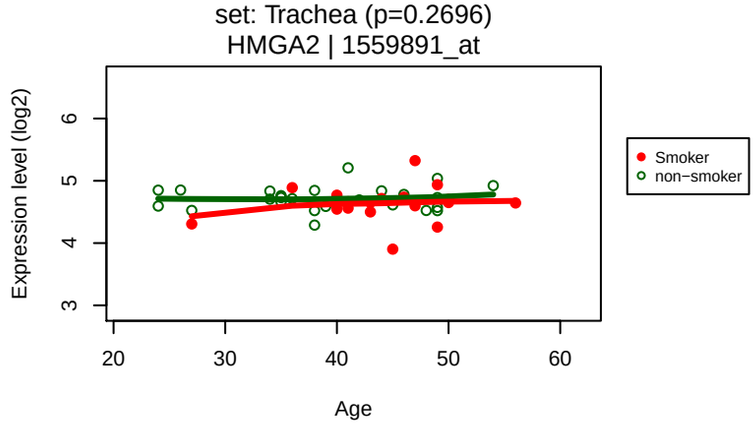

Gene expression ratio in Males
gene: HMGA2 | 1559891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0773
p=0.7717
p=0.4218
p=0.0161
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HMGA2 | 1559891_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4446
p=0.0558
p=0.4293
p=0.4369
SmallAirway2
LargeAirway2
Trachea
Macrophage2

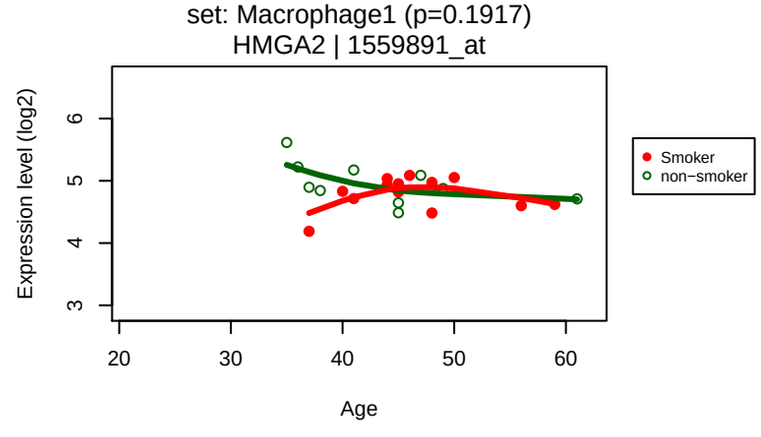

set: SmallAirway1 (p=0.9386)
HMGA2 | 1559891_at
Expression level (log2)
Age
Smoker
non−smoker

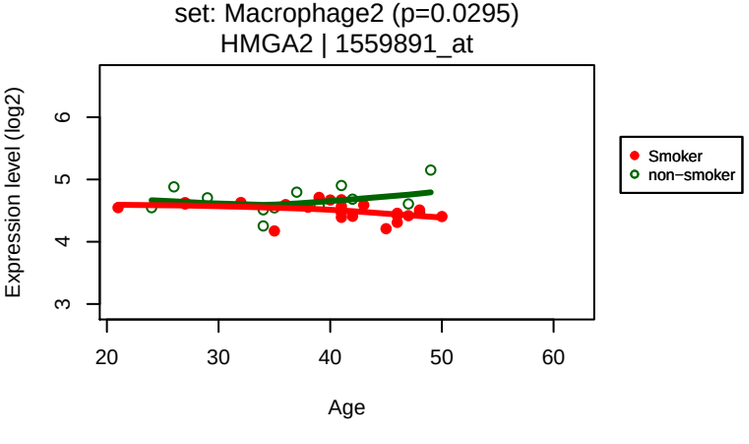

set: SmallAirway2 (p=0.1627)
HMGA2 | 1559891_at
Expression level (log2)
Age
Smoker
non−smoker

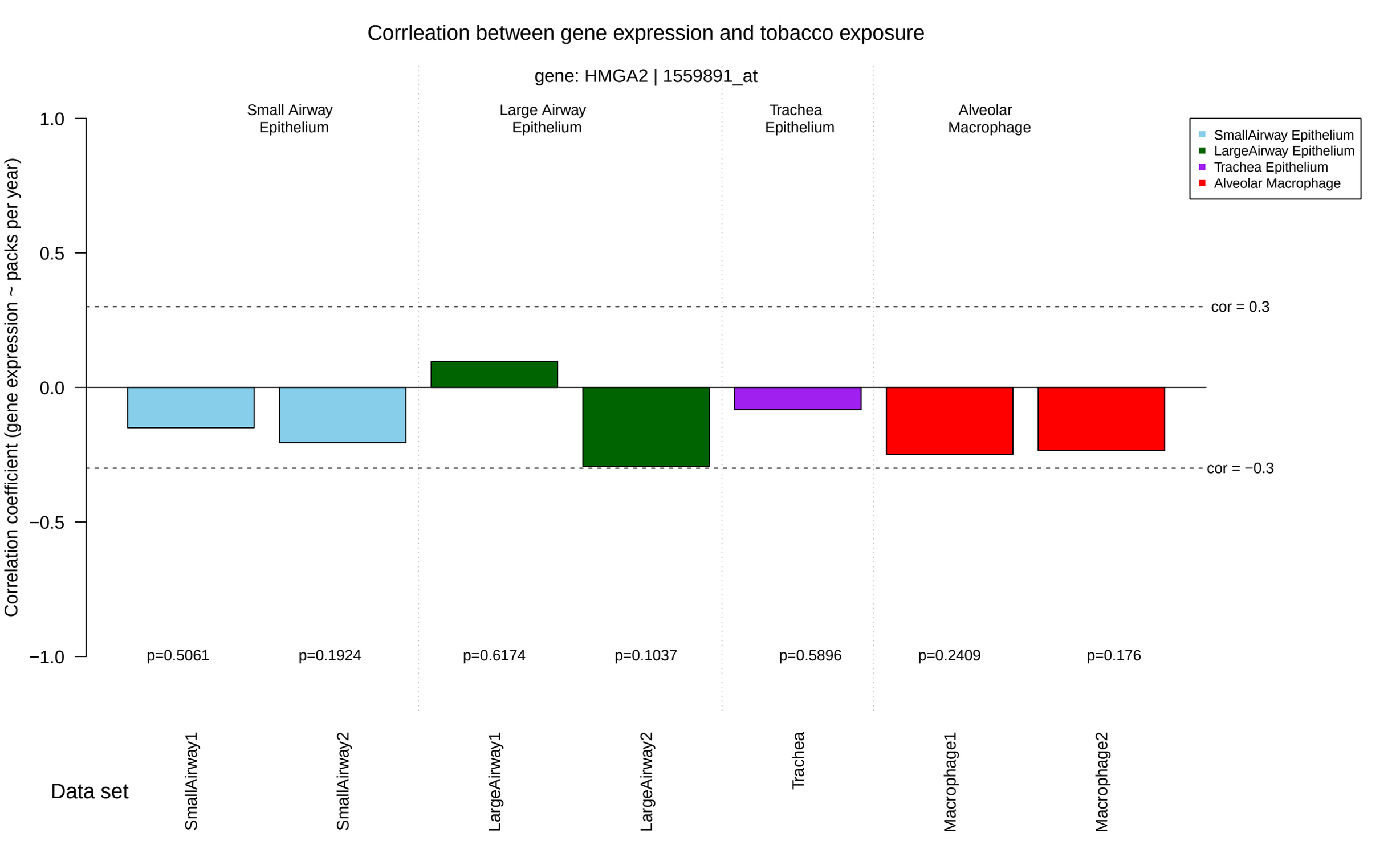

set: LargeAirway1 (p=0.3615)
HMGA2 | 1559891_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4379)
HMGA2 | 1559891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2696)
HMGA2 | 1559891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1917)
HMGA2 | 1559891_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0295)
HMGA2 | 1559891_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HMGA2 | 1559891_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5061
p=0.1924
p=0.6174
p=0.1037
p=0.5896
p=0.2409
p=0.176
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2